CatWISE Catalogue Columns and Flag Filters

This document provides a brief outline of the necessary columns to extract from the CatWISE2020 catalogue to create the smaller dataset used for cross-matching. It also provides more information on the "flag" index, additional columns to extract and combine into a quality flag value.

Each catalogue will need to be transformed from its current state in the DAC to separate files for ingestion into the cross-match process, to allow for a common interface for flexible matching across multiple different catalogues. A summary of this in its current form is provided below.

ID

This is the string `source_name` from the primary catalogue. Alternatively, `source_id` can be used if desired for more efficient internal data table matching.

Sky coordinates, uncertainty

Sky positions are `ra_pm` and `dec_pm` ; note the `pm` suffix, indicating we use the more-complex five-parameter solution. The circular astrometric uncertainty is calculated through the averaging of semi-major and semi-minor axis uncertainties, calculated as

```
sig_maj = sqrt(0.5 * (sigra_pm**2 + sigdec_pm**2 + sqrt((sigra_pm**2 - sigdec_pm**2)**2 +
                                                        4 * covar**2)))
```

and

```
sig_min = sqrt(0.5 * (sigra_pm**2 + sigdec_pm**2 - sqrt((sigra_pm**2 - sigdec_pm**2)**2 +
                                                        4 * covar**2)))
```

with the additional step that `covar` needs to be calculated from the catalogue as well,

```
covar = sigradec_pm * abs(sigradec_pm)
```

with `sigradec_pm` being the *co-sigma* between `ra_pm` and `dec_pm` .

We wish to take the geometric mean (instead of the arithmetic mean), $\left(\prod_{i=1}^{n} \sigma_i\right)^{\left(\frac{1}{n}\right)}$, to preserve the ellipse area, so

```
sig_avg = sqrt(sig_maj * sig_min).
```

Magnitudes

The two magnitudes to be recorded in the table are `w1mpro` and `w2mpro` . Non-detections and non-sources (with `-999` or `-1` flags) should be updated to have `NaN` magnitudes once the flags (below) have been computed.

Signal-to-noise ratios

For CatWISE2020 we can use the directly calculated signal-to-noise ratios columns `w1snr` and `w2snr` . if `w*mag` is `NaN` then the corresponding `w*snr` entry can also be set to `NaN` .

Best detection index

The table for *WISE* flag selection, defaulting to a flag number of `2` unless criteria are met, is given as:

Flag	Flag Number	Criteria
Non-source	-999	`sigra` <= 0 OR `sigdec` <= 0 OR `sigra` == NaN OR `sigdec` == NaN
Non-detection	-1	`ab_flags` is not `0` (being instead `D` , `H` , `O` , or `P`) OR `w*mpro` == NaN OR `w*sigmpro` == NaN OR `w*sigmpro` <= 0 OR `sigra` >= 2 OR `sigdec` >= 2

Flag	Flag Number	Criteria
Low quality (saturated)	0	`w*sat` > 0.1[1]
Low quality (generic)	1	`w*snr` < 5 OR `w*sigmpro` $>= 0.5$

Note that `w*sat`, e.g., applies to each filter in turn: `w1sat` and `w2sat`, while `ab_flags` are two-character strings, with each string applying to a separate filter in turn as `[W1][W2]`, such that the `W1` flag for photometric quality should be read `wise[i]["ab_flags"][0]` (or `ab_flags[1]` for one-indexed languages) for the ith source. Note that if the non-detection or non-source flag for `sigra` or `sigdec` is met (for non-positive or unphysically large uncertainties), then the source should be flagged as "bad" in all filters and essentially discarded. The criteria of `sigra` and `sigdec` being below 2 arcseconds stems from an approximate requirement that the uncertainty be at most a fifth the maximum separation considered for two sources to be potential counterparts; this is currently set at 11 arcseconds, and hence $11 / 5 \simeq 2$ – this has potential issues going forward if CatWISE is matched to a significantly less precise dataset, but for now is likely an acceptable value.

Each source has two filter flags, with lower value flags overwriting any other flags for a given filter. The *WISE* filter priority in wavelength and numerical order: pick `W1` first and `W2` otherwise. In turn, sources are searched for a `2` flag, with ties assigned to the lower number – and smaller catalogue index – filter choice. If no sources have a `2` flag, repeat for a `1` flag, then a `0` flag. As always, sources should not be included in the main catalogue if they only contain negative flags across all four filters, and as such we should always return a `best_flag` from a search of `2` -> `1` -> `0` flagged detections in the four filters.

Epoch and proper motion

For CatWISE2020, the epoch of all recorded positions is an MJD of 57170 (`J2015.405`). However, this is information that we only store here for the purposes of recording the epoch that other catalogues may have to use as their move-to epoch; we do not wish to use proper motions from the CatWISE data, and so do not include any epoch or motion information in our reduced table.

Zone overlap index

Not part of the original catalogue, but during the table-creation process a flag for "is this object in the core of the chunk", as per the core-halo model, will be required.

Reject catalogue

Any object that is removed from the final catalogue should be placed in a separate "reject" catalogue and provided separately. As these will fail to meet the best detection criterion that does not need storing; however, we still will require the zone overlap index.

[1] `w*sat` can be NaN, but as I understand it this can only occur when no pixels are used to fit the source – this simply being the ratio of saturated pixels to total pixels used in the fit for this source. In this case I am assuming therefore that `w*mpro` must also be NaN and the object removed accordingly in this band. However, this may interfere with a naive inequality within the code, so it may be necessary to add a "is not NaN" check as well.